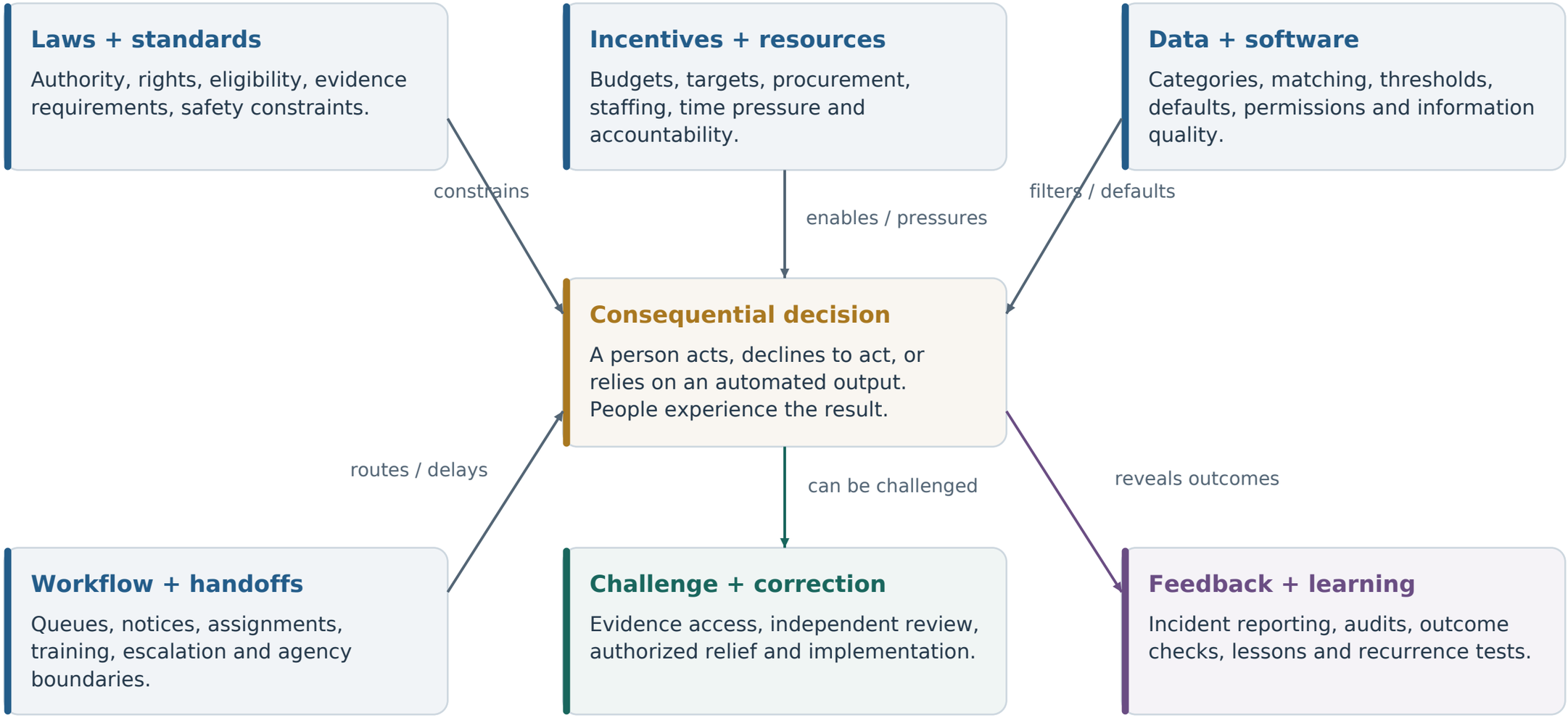
What systems shape human safety behavior?

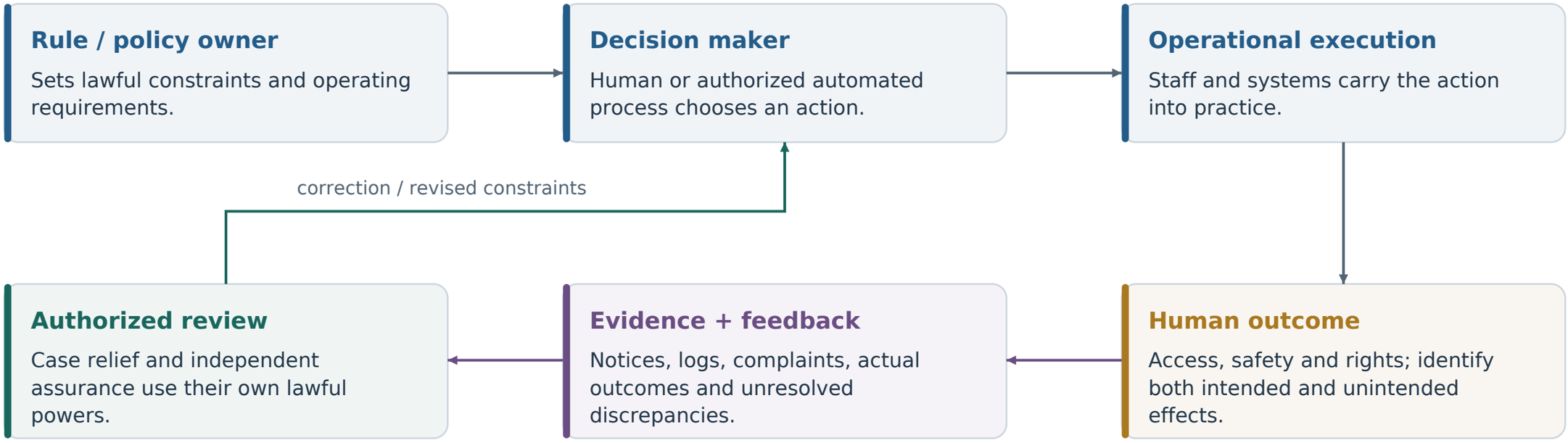
Conceptual grouping by function. Links are mechanisms to investigate, not measured causal effects or statistical clusters.



Inspect the interaction: a well-intentioned person can still be constrained by an unsafe workflow.

Where are the commands, checks and feedback?

A systems-safety model adapted to public decisions. Each control needs an owner, authority, evidence and a test.



FOUR QUESTIONS FROM SYSTEMS SAFETY

Unsafe action

Was an action taken when its safety conditions were absent?

Missing action

Was a required protective action omitted?

Wrong timing

Was action too early, too late or in the wrong order?

Wrong duration

Did action continue too long or stop too soon?

Where do you take a procedural safety defect?

Use the failure type to identify the lawful correction owner. Individual relief and system repair may proceed in parallel.

Failure to correct	Implementation owner	Review / assurance route	Completion evidence
Wrong / stale record	Source program + records / privacy office	Record-review process; judicial review where authorized	Corrected record + recipient confirmations
Unsafe repeated procedure	Agency process owner + internal-control / quality staff	Relevant auditor / IG; rulemaker if formal rule must change	Tested workflow + recurrence monitoring
Automation / matching defect	Program + IT/data + procurement owners	Privacy/security review; Data Integrity Board for covered federal matching	Versioned fix + tested effects on decisions
Harmful individual ruling	Designated hearing / reconsideration authority; court with jurisdiction	Available appeal / writ / stay process	Lawful relief + proof of implementation
Court filing / access barrier	Clerk / court administration; access coordinator where applicable	Presiding judge / appropriate judicial or civil-rights route	Filing/access restored; issue documented
Misconduct / interference	Relevant conduct / disciplinary authority	Judicial conduct body or agency IG within its mandate	Findings + authorized corrective action
Law / funding / contract defect	Legislature; delegated rulemaker; budget / procurement owner	Legislative oversight; audit; available judicial review	Changed requirement + implementation test

Functional routing, not one universal complaint office. Powers and deadlines vary. M01, M04-M08; prior institutional sources E01-E09.

A documented route is not a demonstrated repair.

D = an official route, role or rule was documented. ? = effectiveness or coverage was not assessed here. This is not a safety score.

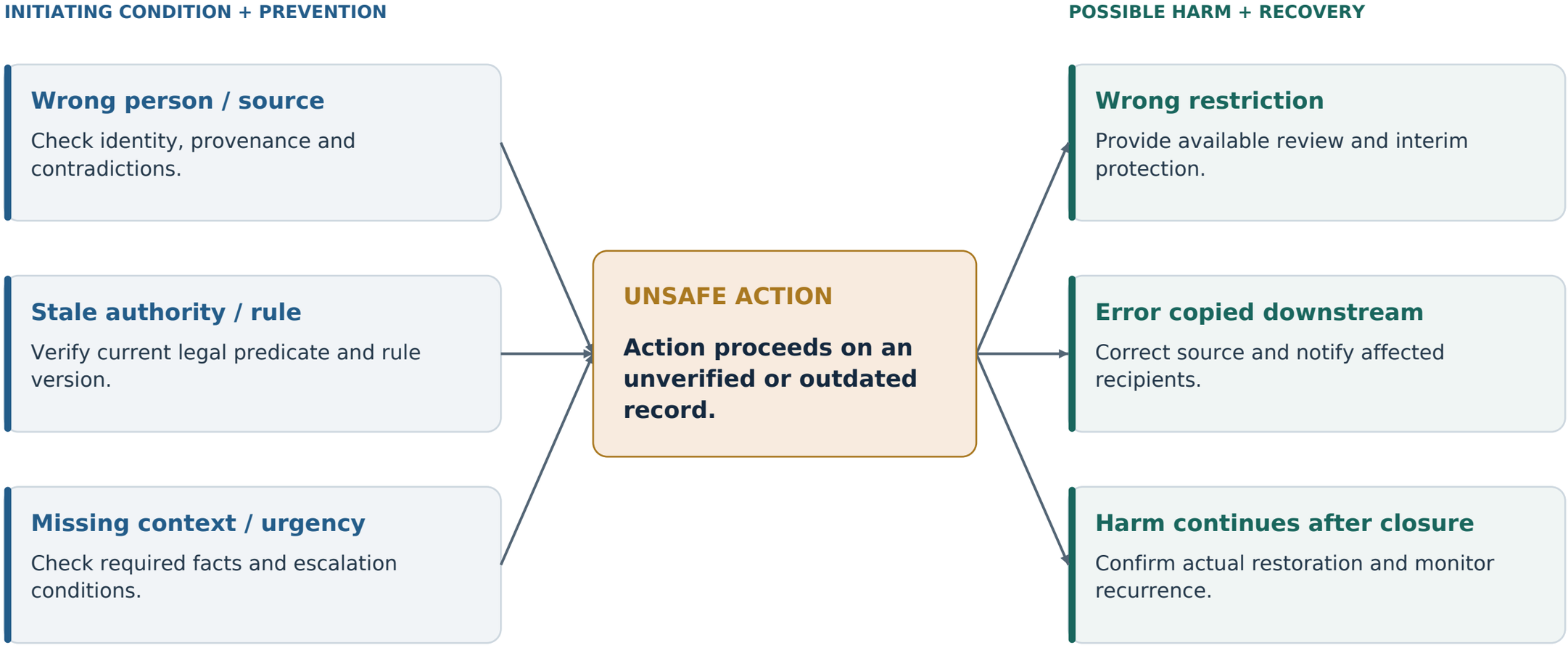
	Review / complaint route	Oversight role	Recipient-correction duty	Timely repair measured	Human outcome verified
Florida DMV	D	D	?	?	?
Courts (OK / CA)	D	D	?	?	?
SSA	D	D	D	?	?
Montana programs	D	?	?	?	?
911	D	D	?	?	?

A gap in this review is a question to investigate

The ? cells do not establish that a safeguard is absent or that harm occurred. SSA’s documented recipient-notification rule does not demonstrate compliance in a particular case.

Prevent unsafe action; interrupt harm; verify recovery.

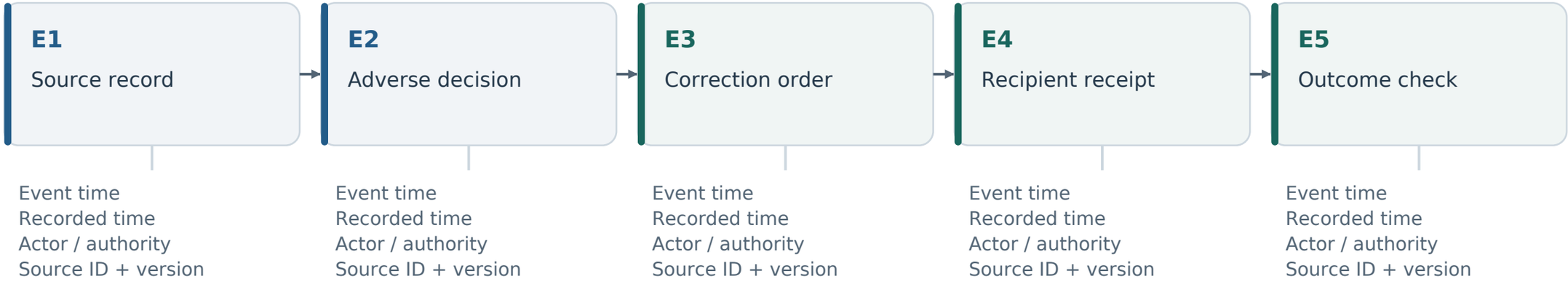
Bow-tie scenario model: an adverse action relies on an unverified or outdated record. No probability or factual allegation is implied.



For each safeguard: owner / trigger / authority / test / evidence / failure response

Preserve the difference between a claim and an event.

Illustrative evidence structure. The event markers below are placeholders, not a chronology of any person’s case.



Provenance / custody

Keep original files, acquisition method, transfer history and integrity checks. A file hash supports integrity after capture; it does not establish that the contents are true.

Contradictions / missing evidence

Retain conflicting versions. Distinguish observed events, reported claims, inferred links, proposed changes and unknowns.

A received correction is not yet a verified outcome

Compare the corrected source, receiving record and actual legal/practical consequence. A sequence of events alone does not prove causation.

M09: NIST digital-evidence preservation. M10: event logs support observed-process analysis; they can omit real activity.

Make safety visible without hiding uncertainty.

Measurement specification for a future case dataset. No results or numerical risk ratings are asserted here.

Question	Measure / test	Interpretation safeguard
Are decisions justified?	Independent sampled review against the applicable authority, evidence and procedure.	Define the reference standard; record reviewer disagreement.
Are interventions mistaken?	False-positive rate: unwarranted interventions among cases where intervention was not warranted.	Requires reliable case labels and a defined population.
Is protection missed?	False-negative rate: missed warranted protective actions among cases requiring them.	Unreported harm and missing cases can bias the estimate.
Does relief arrive in time?	Time from identified need/error to authorized relief and to verified safe state. Report distributions.	Keep event time distinct from record-entry time.
Did the correction travel?	Classify eligible repairs as complete, incomplete or unknown across required known recipients.	Define cohort, cutoff date and completion evidence.
Did the defect recur?	Track repeated failure after a repair over a stated follow-up period and exposure denominator.	Compare case mix and changes in detection, not just counts.

Choose the method to match the evidence.

Charts help test explanations. They cannot turn an unverified claim into a fact or an association into a proven cause.

Visual / method	Useful question	Evidence requirement / limit
Concept / cluster map	Which functions and dependencies shape behavior?	Conceptual grouping is available now. Statistical clustering needs coded observations and a justified distance measure.
Safety control map	Who can act, what constrains action, and what feedback reaches them?	Verify each authority, control action and feedback channel. A model may omit relationships.
Coverage heatmap	Which safeguards have documentation and which have been tested?	Distinguish documented, tested, failed, unknown and not applicable. Do not manufacture scores.
Hazard / safeguard map	Where can an unsafe condition become harm, and what interrupts it?	Label scenarios and controls; test mechanisms before inferring failure probabilities.
Timeline / provenance graph	What happened, what is claimed, and which record supports each link?	Preserve source versions, times and custody. Copies of one assertion are not independent corroboration.
Process mining / flow map	How do real cases move, stall, loop or depart from the required process?	Needs case IDs, activities and timestamps. Log completeness and capture practices constrain conclusions.

Useful existing foundations for procedural safety

These mechanisms have different legal scopes. None is a single universal safety office above all courts and agencies.

Federal internal control

GAO’s Green Book sets internal-control standards for federal executive agencies. Management implements controls. Other governments may adopt the framework; GAO is not a personal appeals tribunal.

Federal records + matching

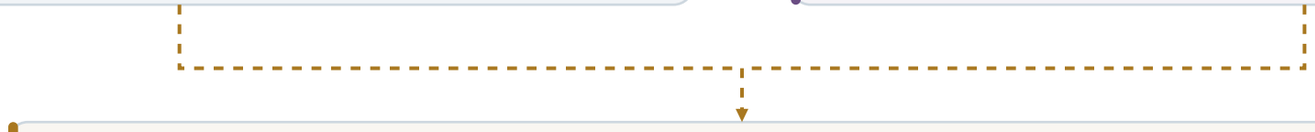
Privacy Act duties cover qualifying records and matching programs. Data Integrity Boards review covered matching agreements and compliance; specified adverse actions have verification / notice safeguards.

Sector-specific safety

FCC regulates relevant 911 provider obligations; state/local authorities operate call centers. Other domains have their own regulators, legal duties and complaint routes.

Case review + independent scrutiny

Hearings and courts can grant assigned forms of relief. Auditors, IGs and conduct bodies perform different oversight functions; the available powers must be verified.



PROPOSED: an assurance coordination layer

Link a defect report to the authorized correction owners. Track individual relief, process repair, recipient updates and verified outcomes. Compulsory powers require a lawful grant of authority.

Existing mechanisms: M01, M04-M08, E01-E09. Proposed coordination is a design suggestion, not an existing universal mandate.

Source directory 1

Official and primary method sources. The diagrams are bounded analytical syntheses, not case-specific findings.

M01 GAO Green Book: federal internal-control standards

https://www.gao.gov/greenbook

M02 ICAO guidance on STPA: safety control structures and unsafe actions

https://www.icao.int/sites/default/files/SMI/TrainingDocs/Chapter%202%20Safety%20Management%20Fundamentals/2.6-05-SRM-Methodology-STPA.pdf

M03 NIST AI RMF Core: Govern, Map, Measure, Manage (voluntary, AI-focused)

https://airc.nist.gov/airmf-resources/airmf/5-sec-core/

M04 5 USC 552a: records, correction, covered matching and Data Integrity Boards

https://uscode.house.gov/us/usc/119-118/t5/s552a/

M05 5 USC 404: Inspector General duties

https://www.govinfo.gov/link/uscode/5/404

M06 5 USC 553: federal rulemaking and petition right, within its scope

https://www.govinfo.gov/link/uscode/5/553

M07 DOJ: ADA Title II regulations for state/local public entities

https://www.ada.gov/law-and-regs/regulations/title-ii-2010-regulations/

Source directory 2

Official and primary method sources. The diagrams are bounded analytical syntheses, not case-specific findings.

M08 CRS Congressional Oversight Manual

https://www.congress.gov/crs_external_products/RL/PDF/RL30240/RL30240.39.pdf

M09 NISTIR 8387: Digital Evidence Preservation

https://doi.org/10.6028/NIST.IR.8387

M10 Process Mining: Event Data

https://www.processmining.org/event-data.html

E01 FLHSMV Inspector General: BAR audit

https://www.flhsmv.gov/pdf/igoffice/202324-18.pdf

E02 Florida Statutes 322.31: licensing judicial review

https://www.leg.state.fl.us/statutes/index.cfm?App_mode=Display_Statute&URL=0300-0399/0322/Sections/0322.31.html

E03 California Judicial Performance: complaint powers and limits

https://cjp.ca.gov/file_a_complaint/

E04 Oklahoma Council on Judicial Complaints: investigatory role

https://oklahoma.gov/cojc/about.html

Source directory 3

Official and primary method sources. The diagrams are bounded analytical syntheses, not case-specific findings.

E05 SSA: benefits appeals

https://www.ssa.gov/apply/appeal-decision-we-made

E06 SSA: 20 CFR 401.65, correction and known-recipient notification

https://www.ssa.gov/OP_Home/cfr20/401/401-0065.htm

E07 SSA Office of Inspector General: functions

https://oig.ssa.gov/about-oig/

E08 Montana: program hearings and review

https://dli.mt.gov/hearings/faqs/unemployment-insurance-faq

E09 National 911 Program: division of federal / local responsibilities

https://www.911.gov/about/

E10 National 911 Program: FAQ and local complaint routing

https://www.911.gov/calling-911/frequently-asked-questions/